Problem Pack: Square Roots and Square Applications from Mathematics 8 Unit 2 Review

Question 6

Problem: 6. Complete each inequality with the two consecutive whole numbers that the square root lies between. a) < √40 < b) < √55 < c) < √110 <

Student Work:

6 < √40 < 7 7 < √55 < 8 10 < √110 < 11

Feedback

Question 7

Problem: 7. Approximate each square root to the nearest tenth. Use nearby benchmarks or nearby tenths to justify your answer. a) √45 b) √83 c) -√72

Student Work:

6 < √45 < 7 √45 9 < √83 < 10 √83 −8.5 √72 , -√72

Feedback

Question 8

Problem: 8. A square garden has an area of 52^2. a) Write the exact side length. b) State whether the side length is rational or irrational. c) Approximate the side length to the nearest tenth. d) State the two consecutive integers it lies between.

Student Work:

a) s = √52

b)

c) √52

d) 7 < √52 < 8

Feedback

Question 9

Problem: 9. Find the area of a square with each side length. a) 5 cm b) 1.2 m c) 3/4 m d) √6

Student Work:

a) 25^2
b) 1.44^2
c) 9/16^2
d) √6^2

Feedback

In question 9, your units for part a and part d do not match the units given in the problem. Remember to keep the same units of measurement as the original question when you find the area.

Question 10

Problem: 10. Move between the measures of each square. a) A square has a perimeter of 28 cm. Find its side length and its area. b) A square has an area of 45^2. Find its exact side length and its exact perimeter.

Student Work:

28 ÷ 4 = 7 s = 7 7^2 = 49 A = 49^2 s = √45 P = 4s = 4 x √45

Feedback

Question 11

Problem: 11. The figure is made from two squares. Use each area to determine the side lengths, then find the perimeter of the outside edge only. 196^2 64^2 shared edge a) Find both side lengths. b) Identify the shared side that is not counted on the outside. c) Find the perimeter of the entire shape.

Student Work:

a) 14
b)
c) 72

Feedback

In question 11, you left part b blank and your final perimeter for part c is not quite right. Try finding the side lengths of both squares first, then add up only the outer edges that make the outside boundary of the whole shape.